Fake News & Misinformation Detection System

Seminar Project Team

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Problem Statement

- Fake news spreads quickly and overwhelms manual verification workflows.
- Binary classification alone is not enough for reviewer trust.
- A practical system must combine accuracy, explanation, evidence, and deployability.

Objectives

- Build a reproducible baseline and a stronger transformer model.
- Deploy the model through FastAPI and Streamlit.
- Add an agentic layer for evidence retrieval, decision synthesis, and uncertainty handling.
- Address monitoring, privacy, robustness, and ethics.

Dataset

- Local FakeNewsNet-style CSV configured in `configs/dataset.yaml`.
- Fields: `text`, `label`; labels: fake and real.
- Stratified split: 70% train, 15% validation, 15% test.

Preprocessing

- Remove HTML and URLs
- Lowercase text
- Normalize whitespace
- Remove special characters

Baseline Model

- TF-IDF with unigram and bigram features.
- Logistic Regression with balanced class weights.
- Strong benchmark and sanity check for the pipeline.

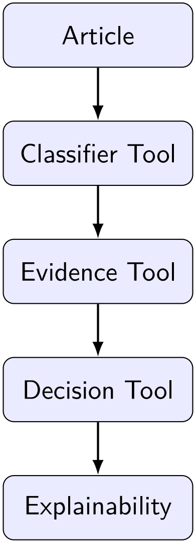
Metric	Value
Accuracy	0.9913
Precision	0.9938
Recall	0.9870
F1	0.9904

RoBERTa Model

- `roberta-base` fine-tuned for binary classification.
- Max length 512, batch size 16, learning rate 2×10^{-5}, 3 epochs.
- Best checkpoint selected by validation F1.

Metric	Value
Accuracy	0.9991
Precision	0.9987
Recall	0.9997
F1	0.9992

Agentic AI Architecture



Evidence Retrieval

- Hybrid HTML-based retrieval improves demo robustness without requiring a single paid API.
- Sources are scored by relevance, credibility, stance, and matched terms.
- Aggregated outputs: support score, contradiction score, quality score, conflict flag.

Explainability

- Final explanation includes prediction, token importance, evidence summary, and review state.
- Trust score comes from the decision layer and stays consistent in the UI.
- UNCERTAIN is surfaced when confidence or evidence is insufficient.

System Architecture

Missing image: System architecture

Evaluation

- Offline metrics: accuracy, precision, recall, F1, confusion matrix.
- Online-style checks: API tests, agent endpoint tests, trace serialization, monitoring writes.
- Comparison is always against the TF-IDF baseline.

Demo Flow

1. User submits article text in Streamlit.
2. FastAPI masks PII and routes the request.
3. Agent classifies, retrieves evidence, makes a decision, and generates explanation.
4. UI shows prediction, trust, evidence cards, and trace timeline.

Results

- RoBERTa outperforms the baseline on all core metrics.
- Agent layer adds evidence, conflict detection, and explainability without retraining the checkpoint.
- Monitoring and traces improve defense readiness and operational visibility.

Missing image: Confusion matrix

Monitoring and Lifecycle

- Monitoring modules track confidence drift, prediction drift, and model health.
- Agent metrics capture evidence success rate, conflict rate, uncertain rate, and response time.
- Retraining strategy is designed conceptually and partially implemented through monitoring outputs.

Limitations

- Evidence retrieval depends on public web availability.
- Evidence stance analysis is heuristic, not learned.
- Public production deployment still needs stronger observability and rate limiting.

Future Work

- Replace heuristic evidence alignment with stance classification.
- Expand robustness and fairness evaluation across broader domains.
- Add managed observability, CI-based release flow, and reviewer feedback loop.

Conclusion

- The project meets the end-to-end NLP, deployment, and agentic AI requirements.
- RoBERTa provides strong performance; the agent layer provides trust and reviewability.
- The final repository is ready for submission and defense with documented risks.

Q&A

Questions?

Prepared topics: model choice, trust score, UNCERTAIN state, deployment, ethics, monitoring.